

THIMIRA GALHENAGE

Mechanical and Manufacturing Engineering (BSc. Hons. Eng)
(University of Ruhuna)

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Mechanical Engineer with experience in industrial machine design, product development, and automation-oriented mechanical systems. Proficient in SolidWorks and AutoCAD with strong skills in 3D modelling, parametric design, design simulations, and detailed engineering drawings. Experienced in designing custom machinery and components with sound knowledge of materials, manufacturing processes, and GD&T. Known for practical design thinking and cost-effective engineering solutions.

WORK EXPERIENCE

Student Instructor - Faculty of Engineering, University of Ruhuna

2025 Aug - 2025 Nov

- Assisted undergraduate students in understanding engineering drawing fundamentals, CAD practices, and technical drafting techniques.

Trainee Mechanical Engineer - Global Engineering, Ansell Lanka (Pvt) Ltd

2024 Nov - 2025 Mar

- Gained hands-on experience in automation and manufacturing processes, including CAD design, PLC programming, and robot programming.
- Designed an automated parametric general liquid texturing tank model using SolidWorks Equation Manager that automatically updates geometry based on user-defined parameters.
- Developed, fabricated, and installed a flow control system for liquid texturing tanks to address high-pressure issues while maintaining compression ratios.
- Performed line balancing on a production line using stations, buffers, and VFD-controlled conveyors.
- Designed components and programmed PLCs for units in the new Printing Machine project.
- Designed and fabricated a new glove former for Ansell Vietnam by analyzing product requirements.

Trainee Mechanical Engineer - Hayleys Fabric PLC

2023 Oct - 2024 Jan

- Hands-on exposure to industrial maintenance, condition-based monitoring, boiler/utility systems, MTTR/MTBF analysis, and engineering documentation.
- Designed a bucket elevator to increase biomass storage capacity and operational efficiency.
- Designed and fabricated an oil filter machine to clean and improve furnace oil quality.
- Designed an automated hopper weighing system to monitor biomass usage accurately.

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Lean Six Sigma Yellow Belt – The Lean Six Sigma Company (2024)
- Java Application Development using JavaSE – Computing Services Centre, University of Colombo School of Computing (2021)
- Pearson Assured Diploma in Information Technology (DITEC) – Esoft Metro Campus, Pearson Assured Organization (2020)
- Graphic Design (Distinction Pass) – Infobit Lanka Computer School (2017)

UNDERGRADUATE PROJECTS

- **Dip Line Automation Project (Ansell Textile Lanka) - Final Year Capstone Project**

Developed an automated glove-dipping process with robotic and PLC integration to reduce headcount and enhance productivity, consistency and labour ergonomics. Applied line balancing, custom robot tooling, simulations and pneumatic systems to optimise performance.

- **Smart Weighing Bin Design Project - Final Year Design Project**

Designed an automated weighing bin with load-cell integration, microcontroller control and a feedback-based dispensing mechanism. Developed CAD models and simulations to improve accuracy and flow efficiency for supermarket bulk-food applications.

- **Glove packing automation project (Ansell Lanka) - Rextro University exhibition project**

Collaborated with Ansell to develop a glove-packing automation demonstration for the Rextro University exhibition. Utilized PLC programming, robot and SCARA arm programming, 3D and 2D vision systems and AMR technology to showcase industrial automation solutions.

- **Aluminium Profile Inspection Machine Project**

Developed an automated defect-detection system for aluminium profiles using AI-based image processing. Integrated machine vision, limit switches, and pneumatic actuation to enable automated sorting, improving inspection accuracy and production efficiency.

- **Salinity Barrier Design Project**

Involved in the engineering design of a salinity barrier across a river near Galle. Contributed to structural design considerations, hydraulic analysis, and feasibility review. Developed analytical thinking in water control structures and environmental impact assessment.

- **Automated Bra Cup Moulding Project**

- **Mini Copra Slicer Machine Design Project**

ACHIEVEMENTS AND EXTRA-CURRICULAR ACTIVITIES

- Winner - Techno 2025 CAD Eager Competition, organized by IESL
- Team Leader (Marketing) - Mexcellence 2025, University of Ruhuna
- Media Secretary - IMechE Student Chapter, University of Ruhuna
- Sports Enthusiast - University of Ruhuna

SKILLS

- Mechanical & Design Software - SolidWorks, Autodesk Inventor, AutoCAD, Ansys
- Automation & Analysis Tools - MATLAB, PLC Programming, Robot Programming
- Maintenance & Inspection - Condition-Based Monitoring, Nondestructive Testing (NDT), Vibration Analysis
- Process Improvement & Lean Tools - Lean Principles, Six Sigma, 5S, Kaizen
- Productivity & Documentation - MS Office, Smartsheet, Google Workspace
- Work Management Skills - Problem-Solving, Critical Thinking, Communication, Teamwork, Time Management

REFERENCE

Mr. Gogullan Amirthakulasingam

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Department of Global Engineering,
Ansell Lanka (Pvt) Ltd.
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